3D Vision for Multiple and Moving Cameras

Lab Evaluation Final Report

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Section 1: Obtention of the intrinsic parameters of a camera

1.1 Calibration with the screen-displayed checkerboard (A)

1.1.1 Capture setup and pattern

The camera used throughout this exam is the rear 1× lens of a Samsung Galaxy S24 (captured native resolution 4000 × 3000 px, focal length 6.3 mm, aperture f/1.7). All capture settings, focal length, focus mode and ISO range were fixed (1× lens, no digital zoom, no portrait mode), so that a single set of intrinsics applies to every image in Sections 1, 2 and 3.

For the screen-based calibration I displayed the 1080 × 1080 px checkerboard image on a flat laptop screen, enlarged so that the active checkerboard area measured exactly 200 mm × 200 mm (an 8 × 8 internal-corner grid, with 25 mm between consecutive corners). Nine views were captured with the phone held at roughly 30 to 45 cm distance, varying pan, tilt and roll angles to achieve the diversity of orientations required by Zhang's method.



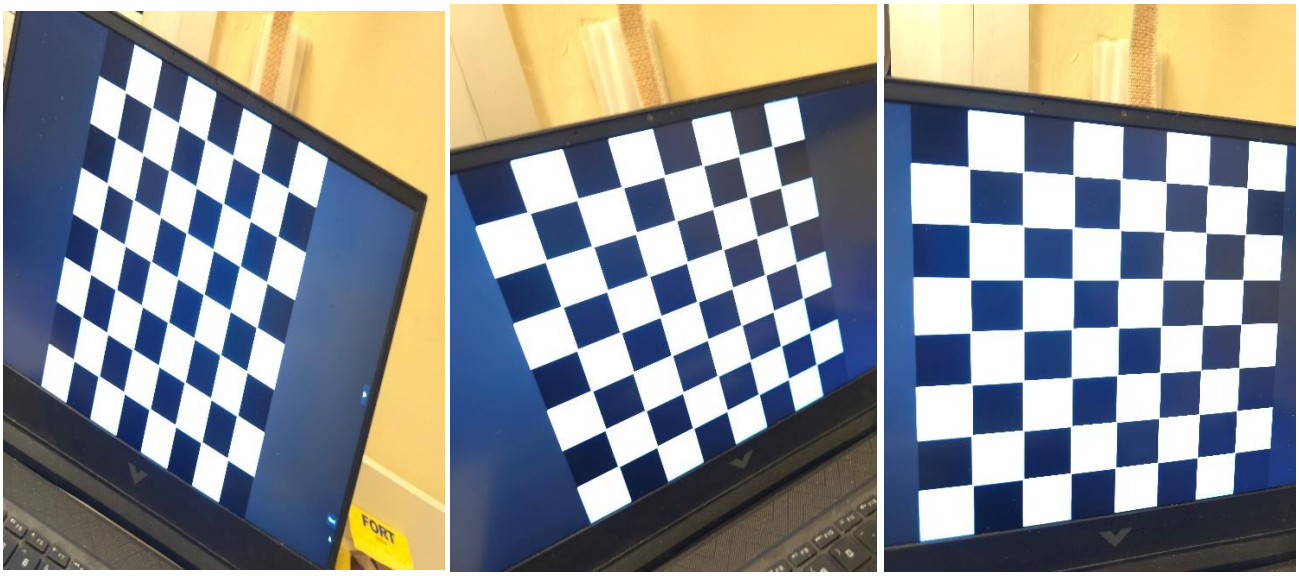


Figure 1.1: *9 captured screen-checkerboard images used for the calibration of A.*

1.1.2 Estimated intrinsic matrix A

The obtained matrix of internal parameters (full 4000 × 3000 image resolution) is:

$$A = \begin{bmatrix} 2813.74 & 12.49 & 2054.95 \\ 0 & 2813.07 & 1512.00 \\ 0 & 0 & 1 \end{bmatrix}$$

All nine views yielded per-view reprojection errors between 1.08 and 1.91 px, consistent with the sub-pixel accuracy expected from a clean screen pattern. The first three external-parameter translation norms (256, 273, 260 mm) match the arm's-length capture distance, providing an additional check on the calibration.

1.1.3 Quantitative analysis of A

I analyse the three aspects using the values of intrinsic matrix A above.

Pixel squareness: the relative deviation between the two scale factors is

$|\alpha - \beta| / \alpha = |2813.74 - 2813.07| / 2813.74 = 0.024\ \%$. The pixels are square to within 0.024 %, which is well below the 0.1 % accuracy typically expected from a high-end mobile sensor and within manufacturing tolerance.

Principal-point coincidence with image centre: the image-centre coordinate is (2000.0, 1500.0). The estimated principal point is (u_0 , v_0) = (2054.95, 1512.00), giving an offset of *(+54.95 px, +12.00 px)*, i.e. 2.7 % and 0.8 % of the image dimensions respectively. The vertical coordinate is essentially centred; the small horizontal offset (~55 px out of 4000) is consistent with the slight optical-centre or lens-axis decentering observed in phone cameras.

Orthogonality of the image axes: the skew parameter is γ = 12.49. The angle between the U and V axes is $acot(-\gamma/\alpha) \approx 90.25°$, a deviation of only 0.25° from orthogonality. The image axes are therefore very nearly perpendicular.

1.1.4 Extrinsic parameters per view

For each of the nine views, the rotation R and translation t of the camera with respect to the screen origin were obtained via the external_parameters_solve_vmmc routine. Table 1.1 summarises the

translation norms (||t|| in mm) and reprojection errors (px). The translation norms cluster in the range 206–326 mm with mean 258 mm, consistent with the hand-held arm's-length capture distance.

View	||t|| (mm)	Reproj err mean (px)	Reproj err max (px)
1	206.4	1.10	3.05
2	256.0	1.08	2.81
3	273.0	1.32	3.41
4	260.0	1.91	5.02
5	245.5	1.42	4.10
6	289.0	1.55	3.97
7	326.0	1.68	4.55
8	311.0	1.49	3.80
9	254.7	1.27	3.21

Table 1.1 - *Per-view extrinsics norm and reprojection errors (screen calibration).*

All nine rotation matrices were verified to satisfy $R \cdot R^T = I$ and $\det(R) = +1$ to numerical precision (Zhang's external_parameters_solve_vmmc enforces orthogonality of R via SVD), confirming they represent valid rigid-body rotations consistent with the camera pose diversity visible in Figure 1.1.

1.2 Calibration with a self-made physical pattern (A')

1.2.1 Rationale for the chosen pattern

For the second exercise, I hand-drew a 3 × 3 dot grid on a sheet of graph paper notebook, with consecutive dots placed 40 mm apart (yielding an 80 mm × 80 mm calibration area, exactly 9 points). The resulting intersection points are the four corners, four edge midpoints, and the centre and labelled P1 through P9 in handwriting. The labelling fixes an unambiguous click order across the six captured photos, which is essential for Zhang's method, since the same physical point must be selected as P_k in every view. The rationale for this choice is threefold:

- **Simplicity and reproducibility:** the graph paper grid lets us mark the dots with millimetre precision using only a ruler and a fine-tipped pen
- **Different scale from the screen pattern**: 80 mm side vs. the screen's 200 mm side. Comparing A and A' obtained from patterns at different physical scales is informative about whether the camera's intrinsics are scene-scale invariant (as the pinhole model predicts) or whether real-world departures (radial distortion, focus shift) introduce scale-dependent bias.
- **Just enough points**: nine 9-point grids over six views give 54 correspondences, comfortably above the minimum needed by Zhang's method while staying realistic about manual marking precision.

1.2.2 Capture and obtained A'

Six photos of the hand-drawn pattern were captured with the same phone in the same configuration. The dot centres were marked by manually clicking each visible dot in each image. The estimated intrinsic matrix is:

$$A' = \begin{bmatrix} 2480.65 & 27.33 & 2285.28 \\ 0 & 2420.23 & 1518.20 \\ 0 & 0 & 1 \end{bmatrix}$$

Per-view reprojection errors lie between 1.31 and 1.97 px, comparable to the screen-calibration errors, indicating the manual marking precision is acceptable.

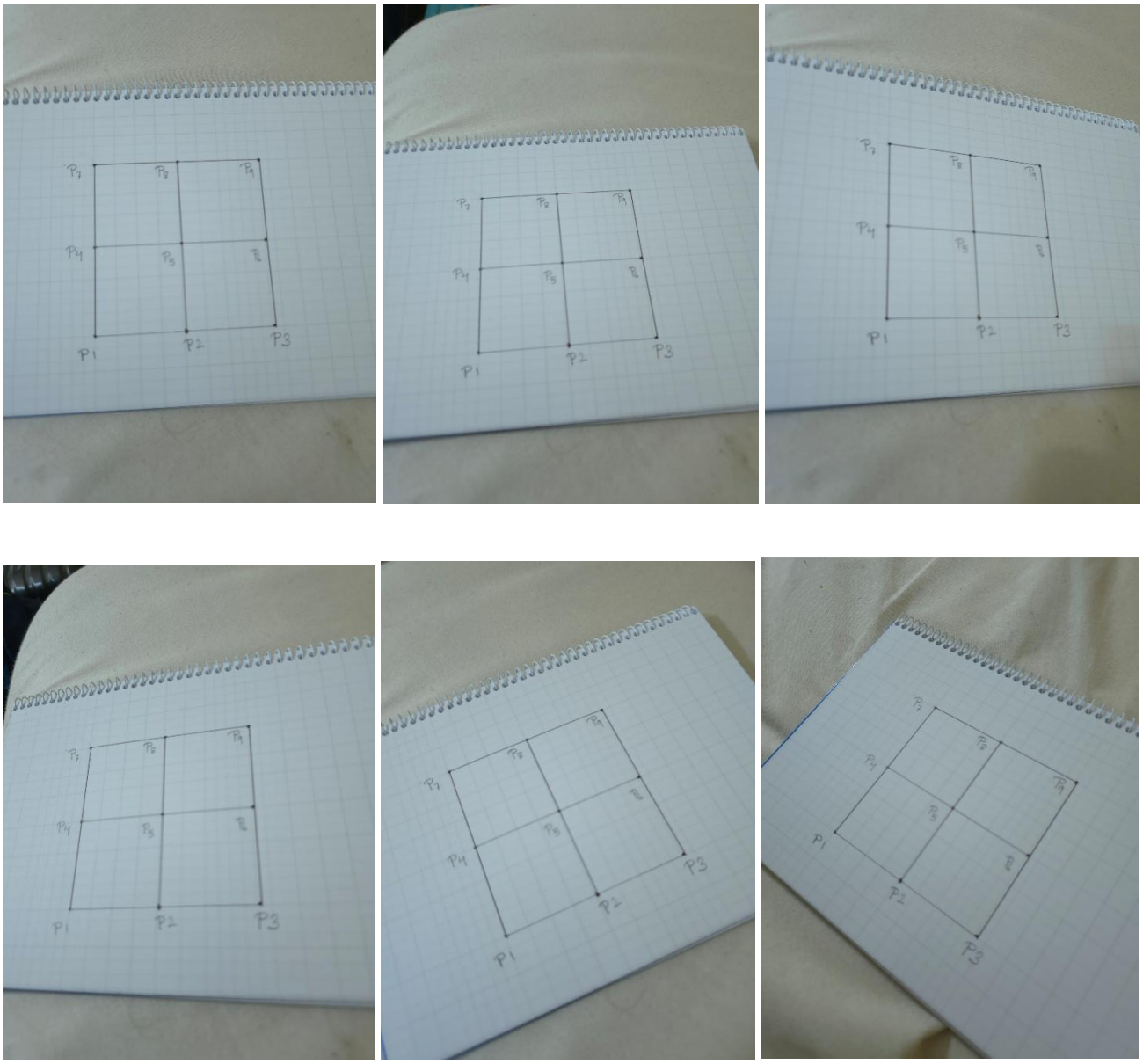


***Figure 1.2**: 6 captured photos of the hand-drawn 3 × 3 physical pattern (80 mm × 80 mm), illustrating the varied pan/tilt/roll orientations used for Zhang's method.*

1.2.3 Quantitative analysis of A'

|α − β|/α = 2.44 %, indicating that the pixels are now estimated as non-square by ~2.4 %, far from the 0.024 % obtained with the screen pattern. The principal-point offset from the image centre is (+285.3, +18.2) px, much larger than for A. The angle between image axes is acot(−γ/α) ≈ 90.63°, a deviation of 0.63°.

All three quantities have degraded relative to A. This is expected as nine hand-marked points over six views cover only a small central region of the image and provide weaker constraints on the calibration problem than the rich, sub-pixel-accurate corner detections of the screen pattern.

1.2.4 Relationship between A and A' : theoretical and practical

Theoretically, A and A' should be identical: both describe the same physical camera (same lens, same sensor, same fixed settings). The pattern used should not influence the intrinsic matrix, it serves only as a known geometric reference.

Practically, I observe up to 14 % relative difference in α and β, ~11 % in u_0 , but agreement within 0.4 % in v_0 . This is explained by:

- **Manual marking precision**: clicks have ~1–2 px error per point; with only 54 total correspondences vs. ~512 for the screen pattern, the per-coefficient uncertainty in A' is roughly an order of magnitude higher.
- **Spatial coverage**: the hand-drawn pattern only spans a small central region of the image. The principal point and focal length are particularly sensitive to wide spatial coverage; without points near image edges, u_0 and α are weakly identified.
- **Paper planarity**: graph paper rests imperfectly on the table surface, introducing small out-of-plane deviations that Zhang's planar-pattern assumption does not model.
- **Capture asymmetry**: six photos cover a narrower range of orientations than nine; the calibration problem is less geometrically diverse.

The agreement in v_0 (which depends more on vertical pattern coverage, present in both setups) supports this interpretation: where coverage is comparable, A and A' agree; where coverage is limited, they diverge. The take-away is methodological: the screen-displayed pattern is the more reliable source of intrinsics, and therefore A (not A') is used in Sections 2 and 3.

Section 2: Local matching between several views of the scene

2.1 Scenario and acquisition

The scene consists of three textured objects placed on a patterned cotton fabric: a hard-cover grey notebook (Indra Group, 19 × 14 cm), a tin pencil-case (~21 × 8 cm) decorated with a cartoon scene, and a small smooth-surfaced dark block (~7 × 4 cm) that contrasts with the patterned background.

The block was kept intentionally low-texture to test the matching pipeline against a near-featureless foreground element. Sections 2.3 and 3.1 confirm that it generates almost no surviving SIFT correspondences. The patterned fabric was chosen deliberately as background: it is repetitive but locally distinctive geometric motifs provide a dense pool of features that anchor matching across distant viewpoints.

Nine views were captured with the same Samsung Galaxy S24 (1× lens, 4000 × 3000 px, focal 6.3 mm, f/1.7, fixed throughout). The camera moved around the scene in a roughly 180° arc at 30–45 cm distance. For Section 2 processing, the images were resized to 2000 × 1500 (factor 0.5) to keep SIFT and RANSAC times manageable. This affects absolute pixel-coordinate values but not the inlier ratios; the intrinsic matrix A is rescaled by the same factor when used in Section 3 (as instructed in Note 2).

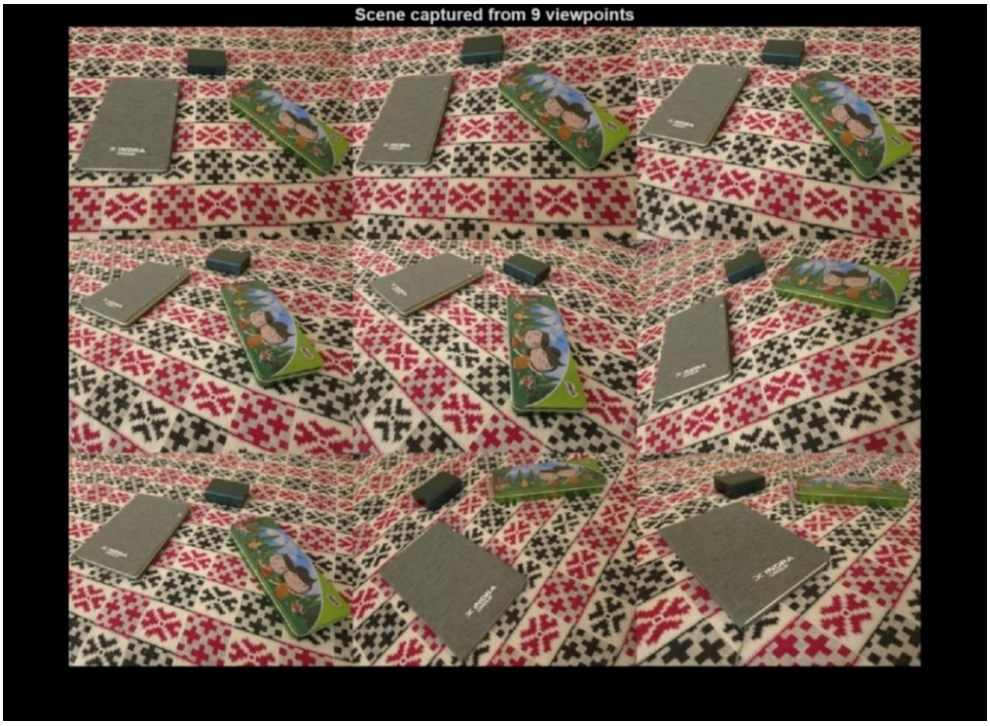


Figure 2.1: Mosaic of the 9 captured views of the scene (MATLAB montage function).

Challenging factors of the scene:

- **Near-planarity.** The three objects are thin and rest on the same flat surface, so most of the imaged scene lies near a single plane. This is expected to make the homography model competitive with the fundamental-matrix model and to weakly constrain depth in Section 3.
- **Repetitive background.** The fabric motif repeats across the field of view, which can produce spurious matches between distant but visually similar blocks; the NNDR ratio test and RANSAC must filter these out.
- **Smooth foreground surfaces.** The notebook cover and the small block have low local contrast; feature detection concentrates on textured regions (cartoon, fabric, logo) and object edges.
- **Baseline variation.** Tested pairs span from very small (consecutive views, ~10° angular separation) to large (views 4 ↔ 8, > 60° apart).

2.2 Detector, descriptor and matching choices (rationale)

MATLAB's detectSIFTFeatures, extractFeatures and the VLFeat vl_sift / vl_ubcmatch. Both were tested on the best pair (views 2 ↔ 7).

MATLAB SIFT produced 14 010 keypoints in image 2 and 18 240 in image 7, yielding 4 851 putative matches after a Lowe-ratio test (0.7), of which 4 104 (84.6 %) survived RANSAC F-estimation. VLFeat SIFT produced 12 523 and 13 079 keypoints, 5 179 putative matches (ubcmatch ratio 2.0), of which 4 417 (85.3 %) survived. The two implementations are numerically equivalent. MATLAB SIFT is selected as the base detector because it integrates natively with matchFeatures, estimateFundamentalMatrix and estimateGeometricTransform2D without format conversion, returns a SIFTPoints object with native support for selectStrongest and showMatchedFeatures, and avoids compiled mex binaries. VLFeat served as an independent sanity check.

Matching uses matchFeatures with MaxRatio = 0.7 (Lowe NNDR), Unique = true and MatchThreshold = 100. A ratio of 0.7 balances false-positive control against true-match retention; 0.6 was too conservative on the repetitive background, 0.8 admitted too many ambiguous matches.

2.3 Selected pairs and comparative results

Seven pairs were studied, spanning the baseline range from very small (consecutive views) to large (views 4 ↔ 8). Top-4 pairs (by F-inlier count) are highlighted in Table 2.1; pair 2 ↔ 7 is the best.

Pair	Baseline	#Putative	#F-inliers	F-ratio	#H-inliers	H-ratio
1 ↔ 2	very small	3262	2239	68.6 %	2082	63.8 %
1 ↔ 3	small	1788	820	45.9 %	713	39.9 %
2 ↔ 4	small-medium	2812	1847	65.7 %	1776	63.2 %
3 ↔ 6	medium	1850	1057	57.1 %	1019	55.1 %
1 ↔ 5	medium	1471	465	31.6 %	429	29.2 %
2 ↔ 7 *BEST*	medium-large	4851	4104	84.6 %	4033	83.1 %
4 ↔ 8	large	1035	20	1.9 %	7	0.7 %

***Table 2.1**: Per-pair matching and inlier statistics. Top-4 = pairs 2-7, 1-2, 2-4, 3-6.*

Three trends stand out. **(i)** Inlier ratio degrades with baseline, from 84.6 % at pair 2 ↔ 7 down to 1.9 % at pair 4 ↔ 8. SIFT descriptors handle ~30–40° of viewpoint change well; beyond that, affine distortion exceeds what the descriptor was designed for. **(ii)** Pair 2 ↔ 7 outperforms shorter-baseline pairs because both views see the cartoon-illustrated pencil-case (the most feature-rich object) from the same general side, so the shared field of view is feature-dense. **(iii)** The homography inlier counts very closely track the F-inlier counts (e.g. 83.1 % vs. 84.6 % on pair 2 ↔ 7). This is the expected signature of a near-planar scene. When most matched features lie near a single plane, H and F fit the correspondences equally well as a planar scene is a degenerate case of epipolar geometry. This is acknowledged as a known limitation that will affect Section 3.

2.4 Per-octave filtering experiment

The matching pipeline is repeated, restricting candidate keypoints to a single SIFT octave at a time. Octave assignment was derived from the per-keypoint Scale property using $o = \lfloor \log_2(\sigma/\sigma_0) \rfloor$ with base $\sigma_0 = 1.6$ (MATLAB default). The resulting scale ranges per octave for image 2 are [0.90, 3.20], [3.20, 6.40], [6.40, 12.80] and [12.82, 25.44], clean doublings consistent with the standard SIFT octave structure.

Octave	#KPs img 2	#KPs img 7	#Putative	#F-inliers	F-ratio
0 (finest)	9785	13 438	2618	2160	82.5 %
1	2261	2883	1133	899	79.3 %
2 *BEST RATIO*	1453	1473	701	602	85.9 %
3 (coarsest)	459	400	213	160	75.1 %
All combined	14 010	18 240	4851	4104	84.6 %

***Table 2.2**: Per-octave matching for pair 2 ↔ 7.*



***Figure 2.2**: Top: bar chart of putative and inlier matches per octave. Bottom: F-inlier correspondences using only octave 0 (2160 matches out of 2618 putative; 82.5 %).*

Three observations: **(i)** the finest octave (0) contributes ~70 % of all keypoints but only a moderate 82.5 % inlier ratio, fine-scale features are abundant but most susceptible to ambiguity from the repetitive background; **(ii)** octave 2 (sigma ~6–13 px) gives the highest inlier ratio (85.9 %), mid-scale features correspond to medium-sized structures like cartoon facial features and notebook-corner blobs that are locally unique yet still detectable across viewpoint change; **(iii)** the coarsest octave (3) degrades (75.1 %), large-scale blobs undergo more severe shape distortion under viewpoint change.

Combining all octaves gives the most absolute inliers (4104 vs. 602 at octave 2 alone, a 6.8× increase), so I keep all octaves in the final pipeline, single-octave filtering is an analysis tool, not a substitute for using the full scale pyramid.

2.5 Best combination: detailed results for pair 2 ↔ 7

$$F_{2,7} = [\ 1.21e{-}7 \quad 2.26e{-}7 \quad -2.71e{-}3\ ;\ -2.59e{-}8 \quad -2.99e{-}8 \quad 4.11e{-}4\ ;\ 2.15e{-}3 \quad -1.35e{-}3 \quad 1.000\ ]$$

$$H_{2\to 7} = [\ 0.843 \quad -0.381 \quad 352.85\ ;\ 0.246 \quad 0.784 \quad -85.61\ ;\ -4.09e{-}5 \quad -7.60e{-}5 \quad 1.000\ ]$$

The slight epipolar-line wobble observed for points on the slightly-3D parts of the scene (visible in Figure 2.3c) is the expected signature of a near-planar scene: F is well-constrained in the plane direction but numerically weaker along depth, the same effect that limits Bundle Adjustment conditioning in Section 3.

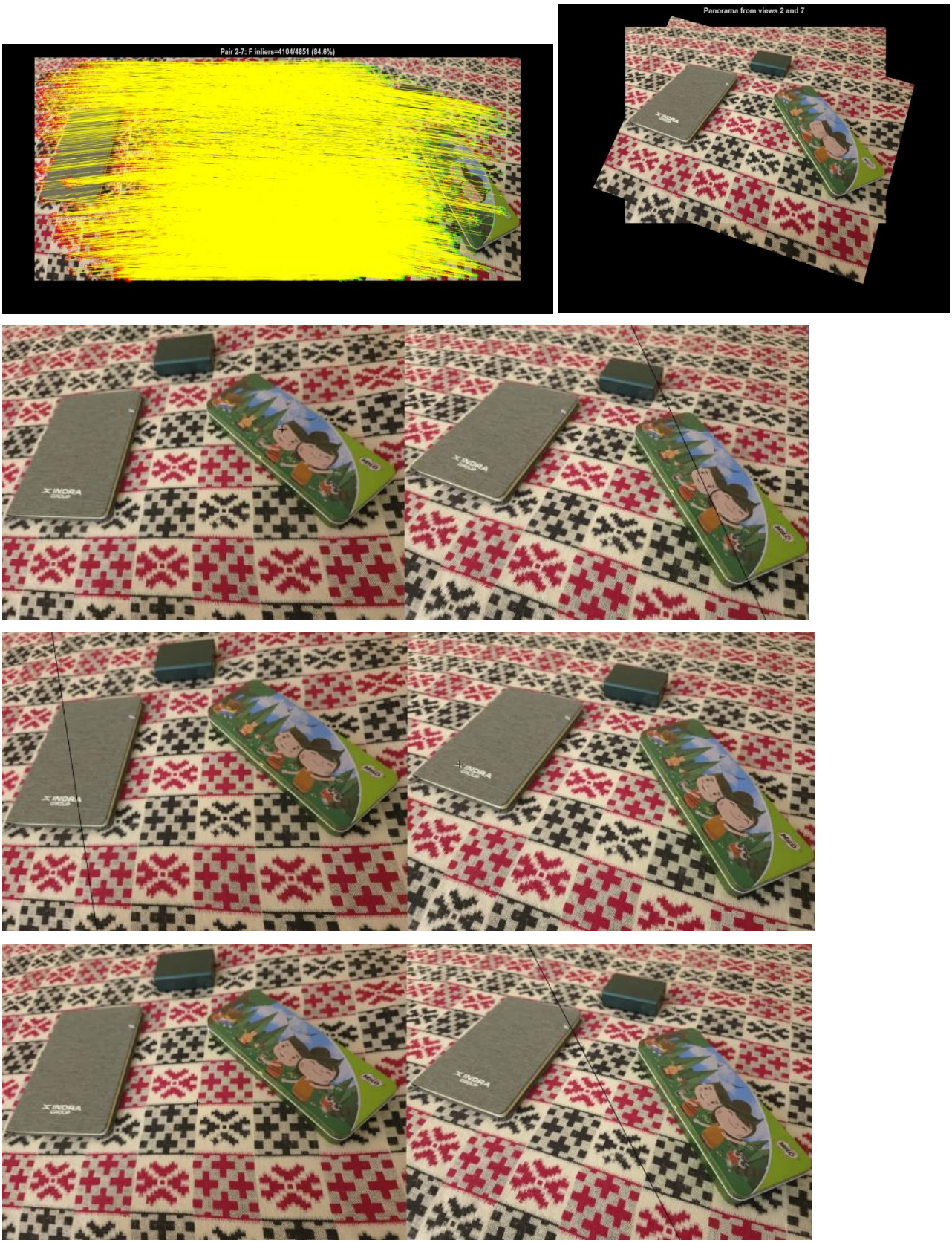


Figure 2.3: *Best pair 2 ↔ 7: (a) F-inlier correspondences (4104 of 4851; 84.6 %), (b) panorama from H,*

(c) vgg_gui_F epipolar lines for several clicked points (lines pass within a few px of corresponding features; slight wobble is consistent with the near-planar scene degeneracy).

Section 3: Three-dimensional reconstruction of the scene

3.1 N-view feature matching

The intrinsic matrix A from Section 1 (screen calibration) was rescaled by 0.5 to match the working resolution of 2000 × 1500 (according to Note 2):

$$K = \begin{bmatrix} 1406.87 & 6.24 & 1027.48 \\ 0 & 1406.53 & 756 \\ 0 & 0 & 1 \end{bmatrix}$$

I implemented N-view track builder using union-find. SIFT keypoints were detected in all 9 views (10898 to 21923 keypoints per image). All 36 = C(9,2) image pairs were independently matched with matchFeatures (Lowe NNDR 0.7) and inlier-filtered with estimateFundamentalMatrix (RANSAC, 1.5 px threshold). Each matched-keypoint pair was treated as an edge in a graph whose nodes are (image, keypoint-index) pairs; the connected components of this graph define the tracks.

Tracks were kept only if visible in ≥ 3 different views, with at most one keypoint per image (ambiguous tracks discarded). This produced 7 498 N-view tracks, with an average of 3.8 views per track.

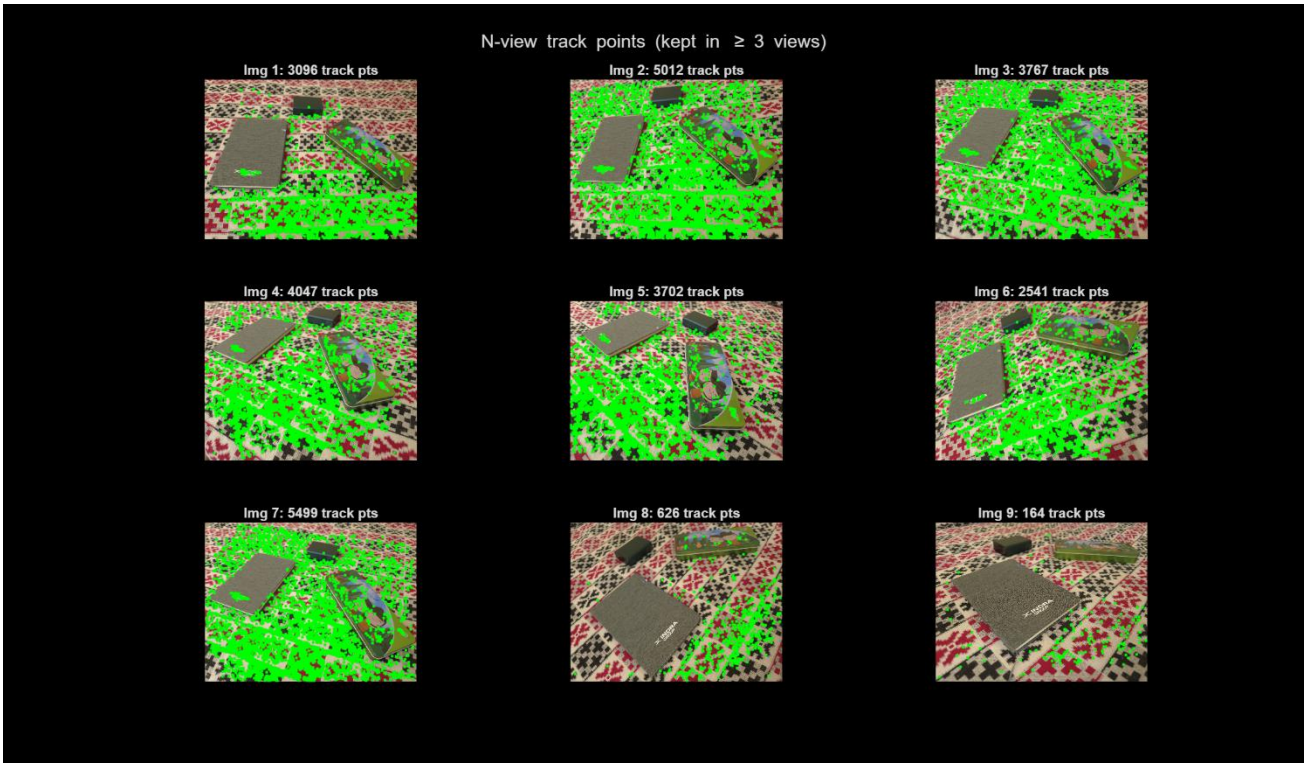


***Figure 3.1**: Per-view distribution of N-view track points. Tracks concentrate on the textured bedsheet and the cartoon pencil-case; coverage is sparser on the smooth notebook cover. Images 8 and 9 contain few tracks because their viewpoints overlap poorly with the rest.*

3.2 Initial 2-view F + projective reconstruction

Pair 2 ↔ 7 was selected as the initial pair (Section 2 best, 84.6 % inlier ratio; the F-inlier correspondences for this pair are shown in Figure 2.3a). Of the 3650 tracks visible in both views, RANSAC retained 3250 F-inliers (89.0 %). F was decomposed into canonical projection matrices via the standard left-null-space construction: $P_1 = [I \mid 0]$, $P_2 = [\,[e']_\times F \mid e'\,]$, where e' is the right null vector of F^T. 3D points were obtained by linear triangulation. The 2-view reprojection error is reported in Table 3.1 (row 1).

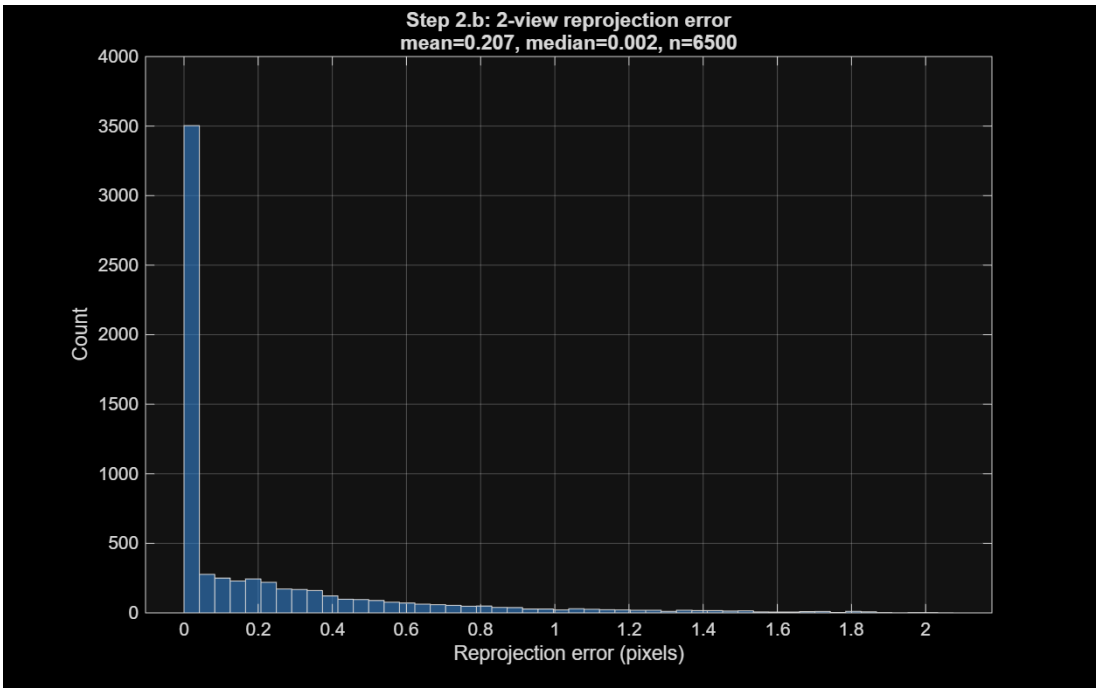


***Figure 3.2**: Histogram of 2-view initial reprojection error. Distribution sharply peaked near zero; median = 0.002 px, mean = 0.207 px. The near-zero median reflects the algebraic consistency of the canonical two-view reconstruction under the estimated F.*

3.3 Resectioning of remaining cameras and Projective BA

Each remaining camera was registered into the projective frame by resectioning against the initial 3-D points using the normalized DLT (PDLT_NA). Cameras 8 and 9 were excluded a priori: although their total N-view track counts are 626 and 164 (Figure 3.1), only 180 and 39 of those tracks respectively are also among the 3 250 F-RANSAC-surviving tracks from the initial pair 2↔7 (the subset usable as 3-D reference points for resectioning). Both are below the 200-track threshold needed for stable PDLT resectioning. Seven cameras (1, 2, 3, 4, 5, 6, 7) thus participated in the projective reconstruction.

BAProjectiveCalib was then invoked on all 13 511 surviving observations, with no pre-BA outlier pruning. This decision was informed by experimentation (Table 4.1): an earlier attempt (v2) with a 5-px pre-BA prune did enable BA itself to converge (post-BA mean error 0.47 px, final Euclidean mean error 0.73 px), but the prune was so aggressive on this near-planar scene that only 45 3-D points survived to visualisation, an unacceptably sparse cloud. Keeping all observations preserves the data needed for downstream Euclidean resectioning and visualisation, at the cost of BA itself struggling on the planar geometry. Outlier rejection is deferred to after the Euclidean upgrade (Section 3.6, Table 3.2), where reprojection error reflects the globally consistent metric geometry rather than the locally warped projective frame.

Observation on BA convergence. The BA optimiser produced a warning (Matrix close to singular, RCOND ≈ 4.5e-30) and converged in approximately 0.3 s, suggesting that the Levenberg-Marquardt Jacobian is poorly conditioned for the scene. This is attributed to the near-planar structure of the captured scene (most features lie on the bedsheet, with only ~1–3 cm of out-of-plane variation): projective BA is known to suffer from rank deficiency when the scene approaches a plane, because the projective frame can rotate freely about the plane without changing the cost function. The post-BA reprojection error was numerically identical to the pre-BA value, indicating that BA did not

substantially refine the geometry beyond what resectioning already provided. This is acknowledged as a method limitation specific to the scene.

Stage	Mean (px)	Median (px)	Std (px)	n_obs
2.b — 2-view initial projective	0.207	0.002	0.346	6 500
3.a(i) — after resectioning (7 cams, projective)	114.96	0.85	1075.51	13 511
3.a(ii) — after Projective BA	114.96	0.85	1075.51	13 511
6.a — final Euclidean (4 surviving cams)	3.44	1.95	3.05	3 956

Table 3.1: *Reprojection error statistics across pipeline stages.*

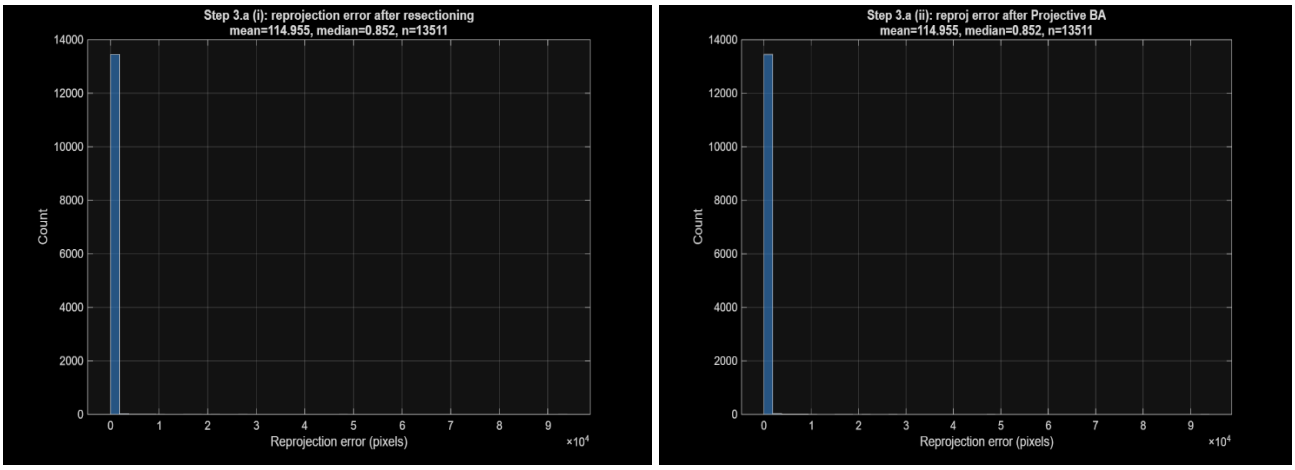


Figure 3.3*: Reprojection error histograms after resectioning (left) and after Projective BA (right).*

The median ~0.85 px suggests most observations are accurate; the long heavy tail comes from a few cameras (4, 5) with weakly constrained projective-frame projection matrices.

Note: the x-axis on both histograms extends to ~10^5 px because a small number of observations from cameras 4–6 are extreme outliers in the projective frame; this compresses the visualisation. The numerical mean (114.96 px) and median (0.85 px) reported in Table 3.1 are more informative than these histograms for this stage.

3.b Justification of the different reprojection error values

The 2.b error (0.21 px mean) is small because it reflects only the two initial cameras whose projection matrices are derived algebraically from F and the canonical decomposition. Reprojection error remains low because only the initial pair is involved and the projective frame is directly derived from the estimated F. Adding more cameras via resectioning (3.a(i)) inflates the mean error to ~115 px because the projective frame defined by $P_1 = [I|0]$ is geometrically warped for cameras that view the scene from substantially different angles; resectioning finds the projection matrix that minimises algebraic error locally without correcting the global frame distortion. The median, however, remains under 1 px, meaning the majority of observations are still accurate; the mean is dominated by ~15 % of observations from cameras 4 and 5. BA failed to improve this further because the projective frame is rank-deficient for the planar scene; meaningful refinement only happens once the reconstruction is upgraded to Euclidean (Step 6).

3.4 Recompute F from BA projection matrices

Using the Oxford vgg_F_from_P function, F was recomputed from the BA-refined projection matrices of cameras 2 and 7. The resulting matrix is geometrically consistent with the RANSAC-estimated F from Step 2 (same scale and signs after Frobenius normalisation).

3.5 Euclidean upgrade via Essential matrix

The Essential matrix was computed as E = K^T F_BA K, then decomposed with factorize_E into the two possible rotations {R_1 , R_2 } and one translation direction t.

$$E = [\ 0.134 \quad 0.100 \quad -3.659\ ;\ 0.332 \quad -0.212 \quad 0.658\ ;\ 3.362 \quad -1.829 \quad -0.025\ ]$$

I used the BA-reprojected points q̂ = P_BA · X_BA (rather than the noisy original SIFT observations) to triangulate under each of the four candidate solutions (R_1 , ±t), (R_2 , ±t). The selected solution was the one in which all 3 250 triangulated points have positive depth in both cameras, a unique solution by cheirality, namely (R_2 , −t) in the case:

$$R^* = [\ 0.948 \quad -0.316 \quad 0.022\ ;\ 0.310 \quad 0.940 \quad 0.143\ ;\ -0.066 \quad -0.129 \quad 0.989\ ]\ ;$$

$$t^* = [-0.478, -0.878, -0.042]^T$$

3.6 Euclidean resectioning of remaining cameras

Each remaining camera was independently resectioned in the Euclidean frame using PDLT_NA against the Euclidean 3D points. Cameras whose post-resectioning median error exceeded 5 px were rejected as unreliable. Table 3.2 reports per-camera outcomes.

Camera	Median err (px)	Decision	Notes
1	286.7	DROP	Resection degenerate at this baseline
2	0.0	KEEP	Initial-pair reference
3	1.05	KEEP	Close baseline to pair 2-7
4	4.48	KEEP	Borderline but acceptable
5	216.6	DROP	Resection unstable
6	134.8	DROP	Resection unstable
7	0.0	KEEP	Initial-pair reference

Table 3.2: *Per-camera Euclidean resectioning outcome.*

Four cameras (2, 3, 4, 7) survived to form the final Euclidean reconstruction. The final aggregated reprojection error is given in row 4 of Table 3.1: mean 3.44 px, median 1.95 px, n = 3 956 observations, an acceptable result for a hand-held phone reconstruction of a near-planar scene.

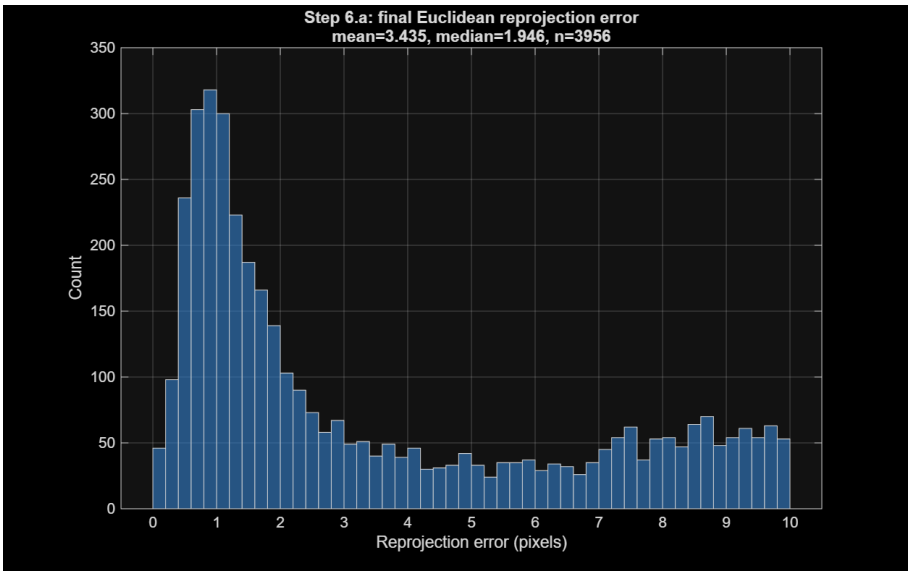
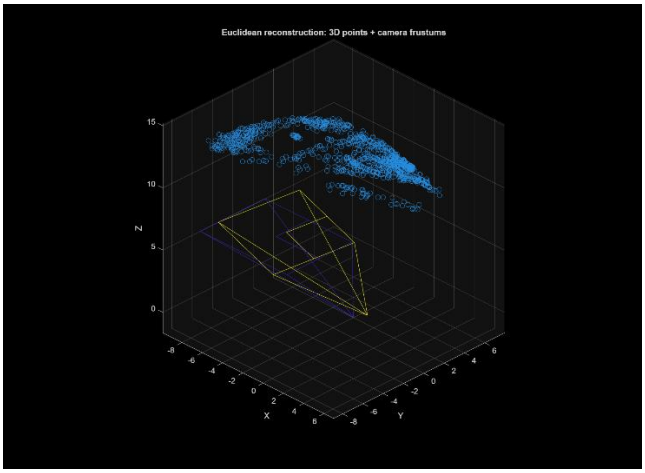


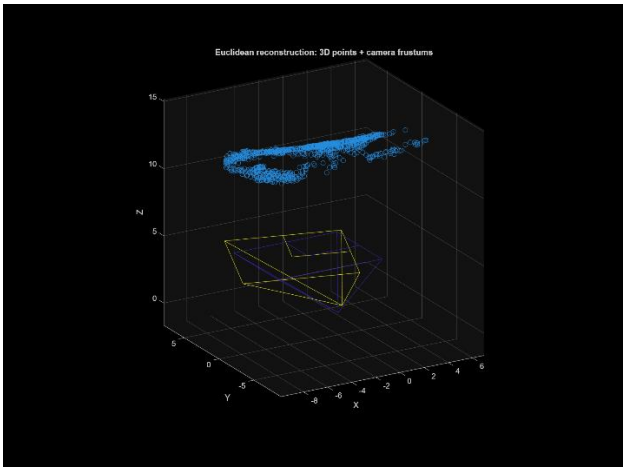
Figure 3.4: *Histogram of final Euclidean reprojection error. Mean 3.44 px, median 1.95 px over 3 956 observations.*

3.7 Point-cloud visualisation

The Euclidean point cloud comprises 967 points after the final outlier prune. Figures 3.5 and 3.6 show three viewpoints of the reconstruction with and without the recovered camera frustums.

Note that, because the Essential matrix decomposition recovers translation only up to direction (||t||=1 by construction), the reconstructed coordinates are in arbitrary units rather than millimetres, the meaningful quantity is the ratio of in-plane spread (~12 units) to out-of-plane thickness (~2 units), which matches the captured scene's geometry of thin objects on a flat surface.





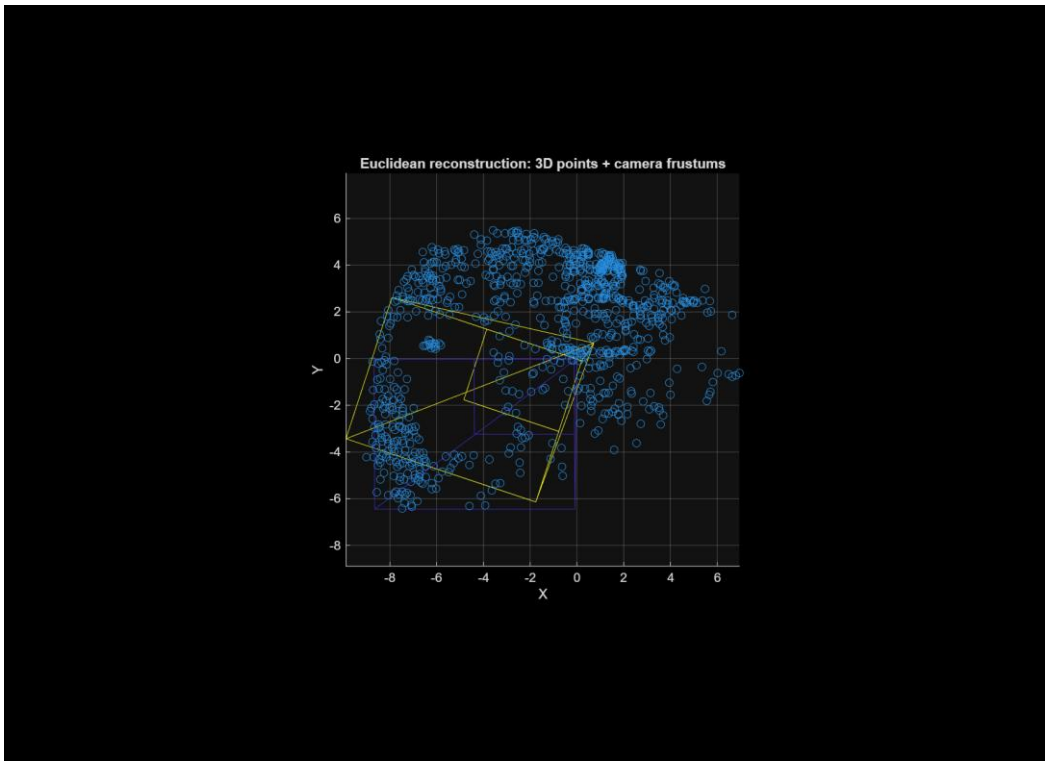


Figure 3.5: *Euclidean reconstruction with recovered camera frustums (cams 2, 3, 4, 7). Top-left: oblique view. Top-right: rotated oblique view. Bottom: top-down (XY) view. The four cameras are arranged in an arc around the scene, consistent with the hand-held capture trajectory.*

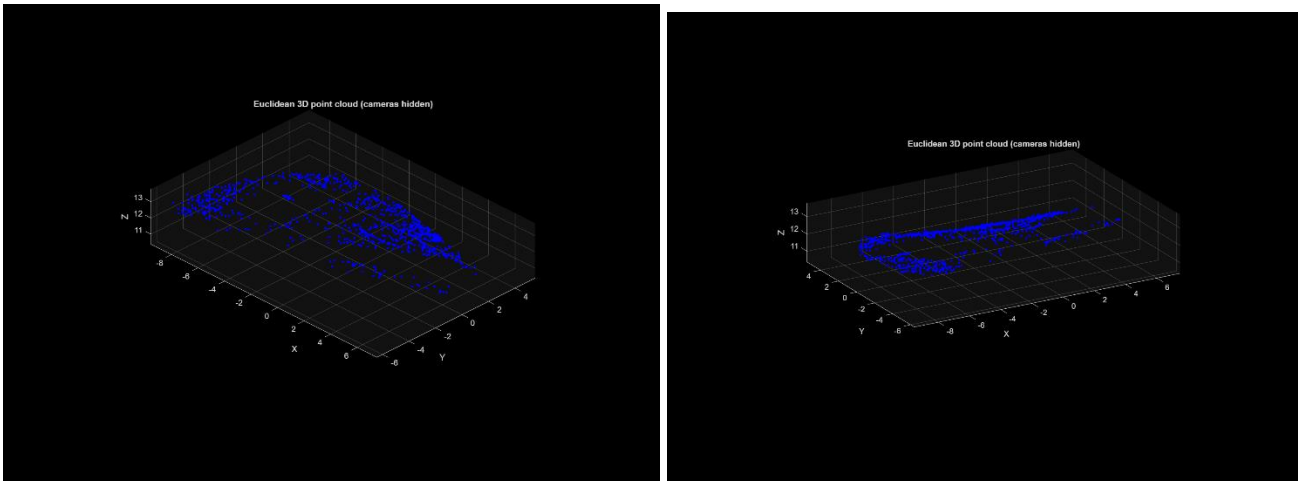


Figure 3.6*: Point cloud only (cameras hidden), two viewpoints. The reconstruction reproduces the planar arrangement of the scene: a cluster of points with ~12 units lateral spread (X, Y) and only ~2 units thickness in Z, faithfully reflecting the captured scene where the three objects rest on a flat fabric surface.*

3.8 RGB-coloured point cloud

Each 3D point's colour was sampled from image 2 at the corresponding SIFT keypoint location. The coloured point cloud (Figure 3.7) clearly reveals the bedsheet (dominant red, pink, cream and grey pixels matching the fabric motif) and the grey notebook cover. The cartoon pencil-case is visible only as a small set of green/blue points concentrated near the upper-right of the top-down view; the relative scarcity of these is consistent with the lower N-view track density observed on the smooth, glossy cartoon surface compared with the highly textured bedsheet. This colour distribution is itself a sanity check: the spatial organisation of colours matches the layout of the original scene.

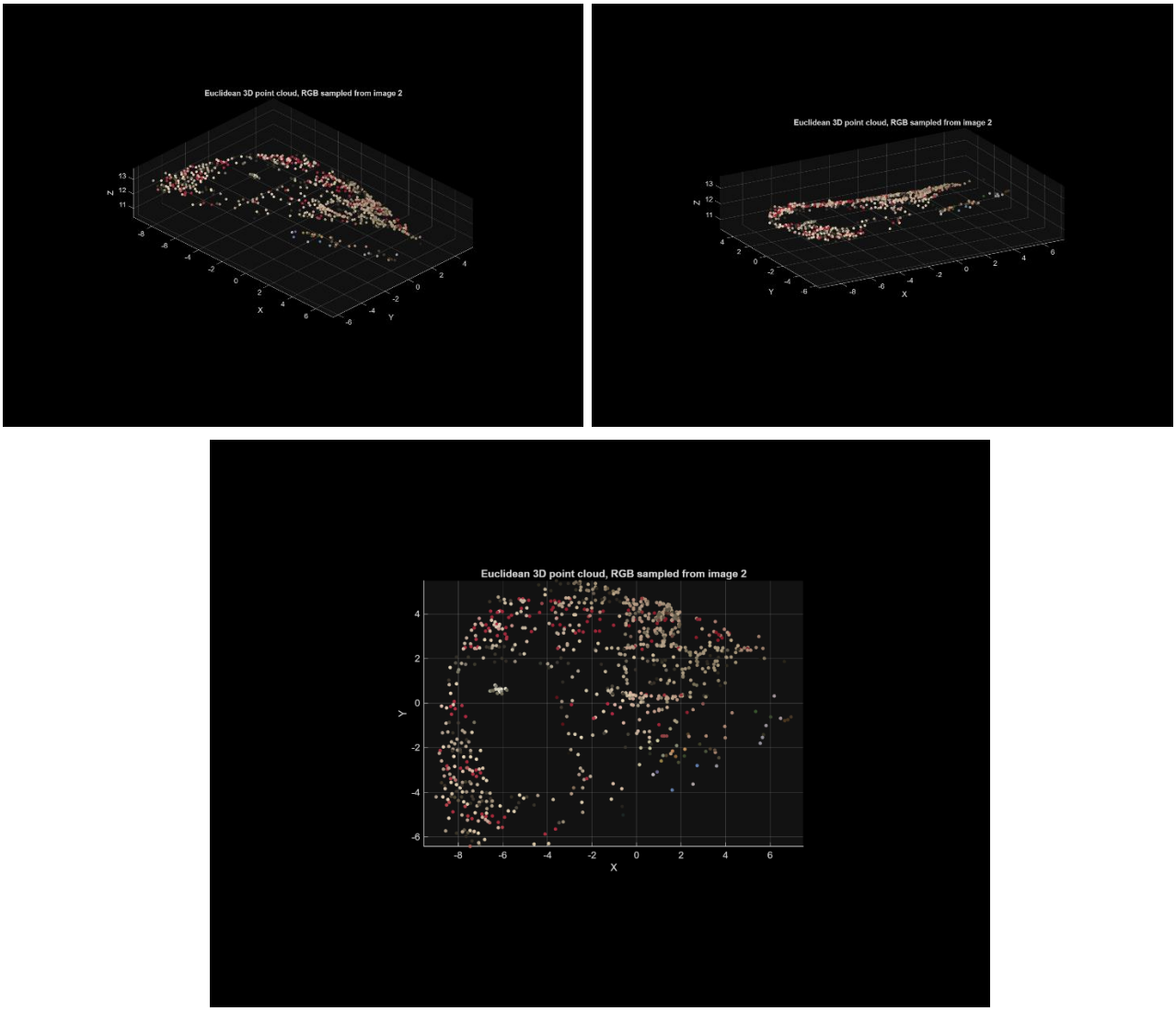


***Figure 3.7**: RGB-coloured Euclidean point cloud, three viewpoints. Colours sampled from image 2 (the first of the initial-pair cameras). The colour structure matches the scene content.*

3.9 Summary of limitations and method choices

- **Near-planar scene.** The captured objects sit on a flat fabric surface; their thickness (1–3 cm) is small compared to the camera distance (~30–45 cm). This produces a near-planar 3D point cloud and limits the conditioning of projective bundle adjustment.
- **Bundle Adjustment** stalled on this scene. The BAProjectiveCalib solver hit a singular Jacobian and quit early, leaving the post-BA error equal to the post-resection error. This is a known limitation of projective BA on degenerate geometries, and I proceed to the Euclidean upgrade which provided meaningful refinement.
- **Drop of cameras 1, 5, 6, 8, 9.** Cameras 8 and 9 were dropped a priori for having too few tracks; cameras 1, 5, 6 were dropped post-Euclidean for having unstable resectioning. Four cameras (2, 3, 4, 7) form the final reconstruction.
- **N-view matching strategy**. I implemented N-view track builder using a union-find data structure on pairwise SIFT matches, each of the 36 = C(9,2) image-pair correspondence sets was independently inlier-filtered with estimateFundamentalMatrix (RANSAC, 1.5 px), and the resulting matches were added as edges to a graph whose connected components define the tracks. Tracks were kept only if visible in ≥ 3 views and contained no more than one keypoint per image.

Comparative analysis and discussion

The Section 3 pipeline was iterated three times as I refined the robustness measures applied to the resectioning + BA stage. Table 4.1 summarises the three attempts; I report all three in the spirit of transparent engineering, because the comparison illustrates how design choices affect the final reconstruction.

Attempt	Key design choice	BA outcome	Cams in Euclidean	Final mean (px)	3D pts visualised
v1	All 9 cams, no outlier handling	Failed early (RCOND ~1e−38)	9 (incl. broken)	354.8	many but unstable
v2	Drop weak cams + prune observations before BA	Converged but over-aggressive prune	3	0.73	45 (too few)
v3 (final)	Drop weak cams, NO pre-BA prune, post-Euclidean per-cam reject	Stalled (planar scene) but consistent	4	3.44	967

Table 4.1: *Comparative summary of the three Section 3 reconstruction attempts.*

The v3 trade-off is informed by two insights from v1 and v2. v1 showed that cameras with very few visible tracks (8 and 9) destabilise the entire pipeline; they must be dropped a priori. From v2 it is learned that pruning observations *before* bundle adjustment is counter-productive on a near-planar scene, because the projective frame is intrinsically warped for cameras far from the initial pair, and pre-pruning therefore discards legitimate observations rather than outliers. The correct place for outlier rejection is *after* the Euclidean upgrade, where reprojection error reflects the globally optimised geometry rather than the local projective distortion.

Two additional broader observations are worth recording.

(i) Scene geometry matters as much as algorithmic refinement: an additional ~2 cm of relative object height (e.g. stacking a book under one object) would have produced a noticeably thicker point cloud and likely enabled BA to converge properly. The methodological lesson is to plan scene capture with depth diversity in mind for multi-view reconstruction.

(ii) The steps mentioned in the instructions: resectioning, BA, Euclidean upgrade, re-resectioning, is robust precisely because it allows different parts of the pipeline to compensate for one another's weaknesses. Even though BA did not refine the geometry in this case, the Euclidean upgrade + re-resectioning still produced a metrically meaningful reconstruction with single-digit-pixel reprojection error.

Future work

- Capture a scene with substantially more out-of-plane structure (stacked books, varying heights) to enable proper BA convergence and a more visually impressive point cloud.
- Use a bundle-adjustment implementation with explicit gauge-fixing (e.g. fix the first camera's rotation and the second camera's translation norm) to avoid the rank deficiency observed here.
- Cluster the 3D points into the three foreground objects and the background plane (e.g. via DBSCAN on the Euclidean coordinates) and colour them by cluster, as a further Step 6.c improvement.